25900219 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZGY101 - BIOLOGICAL BASIS OF BEHAVIOR II

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which structure is found in both prokaryote and eukaryote? [K] / [CO1]

   a). Golgi body b). Nucleus

   c). Ribosome d). Mitochondria

2. During which phase of the cell cycle does DNA replication occurs? [K] / [CO1]

   a). S phase b). G1 phase

   c). G2 phase d). M phase

3. Endoplasmic reticulum with ribosomes are called. [K] / [CO1]

   a). SER b). Rough Endoplasmic reticulum

   c). Lysosomes d). Vesicles

4. Why are sex-linked traits more common in males? [U] / [CO2]

   a). Males have more chromosomes.

   b). Males have only one X chromosome.

   c). Males inherit two X chromosomes.

   d). Males have a stronger immune system.

5. What is the function of a gene? [K] / [CO2]

a). To control cell division To carry genetic information

b). To determine trait

c). To produce energy

d). To protect the cell from viruses

6. Which of the following is a pyrimidine base in DNA? [K] / [CO2]

a). Cytosine b). Adenine

c). Guanine d). Thymine

7. What is a sex-limited trait? [K] / [CO2]

a). Expressed in both sexes b). Expressed only in one sex

c). Only found on the X chromosome d). Only found on the Y chromosome

8. Why do people with Down syndrome have intellectual disabilities? [U] / [CO3]

a). Extra chromosome 21 b). Lack of a specific gene c). Lack of oxygen at birth d). Virus infections

9. Euthenics focuses on improving the quality of life through what? [U] / [CO4]

a). Genetic selection for desirable traits

b). Better nutrition and living conditions

c). Genetic counselling

d). Restricting marriages between affected individuals

10. Which of the following is NOT a common reason for seeking genetic counselling? [An] / [CO4]

a). Family history of genetic disease

b). Difficulty achieving pregnancy

c). Concerns about environmental exposure

d). General healthcare checkup

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. What is cytokinesis? [K] / [CO1]

12. Explain the role of neurotransmitters in cell signalling. [U] / [CO1]

13. Define meiosis. [U] / [CO1]

14. What is monohybrid cross? Give an example. [U] / [CO2]

15. Which are the different sites in a ribosome? [U] / [CO2]

16. Define a reciprocal cross. How does it differ from a normal cross? [U] / [CO2]

17. How many types of gametes are produced in a dihybrid cross? Describe with suitable example. [U] / [CO2]

18. Explain the chromosomal makeup of a person with Klinefelter's syndrome. [U] / [CO3]

19. What is the role of a karyotype in detecting chromosomal abnormalities? [K] / [CO4]

20. Explain somatic mutation. [U] / [CO3]

21. What is frameshift mutation? [U] / [CO3]

22. Explain the role of selective breeding in eugenics. [U] / [CO4]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. What is Cell Communication, and Why is It Important for Living Organisms? [U] / [CO1]

24. Define juxtacrine signalling. Give an example. [K] / [CO1]

25. Briefly explain transcription. [U] / [CO2]

26. Explain Law of Independent Assortment with an example. [U] / [CO2]

27. Discuss the experimental evidence that led to the acceptance of the Chromosome Theory. [U] / [CO2]

28. What is the role of gene mutations in chromosomal aberrations? [K] / [CO3]

29. How can pedigree analysis be used to evaluate the ethical considerations of genetic counseling for a family with a history of haemophilia? [An] / [CO4]

[4x5 = 20]